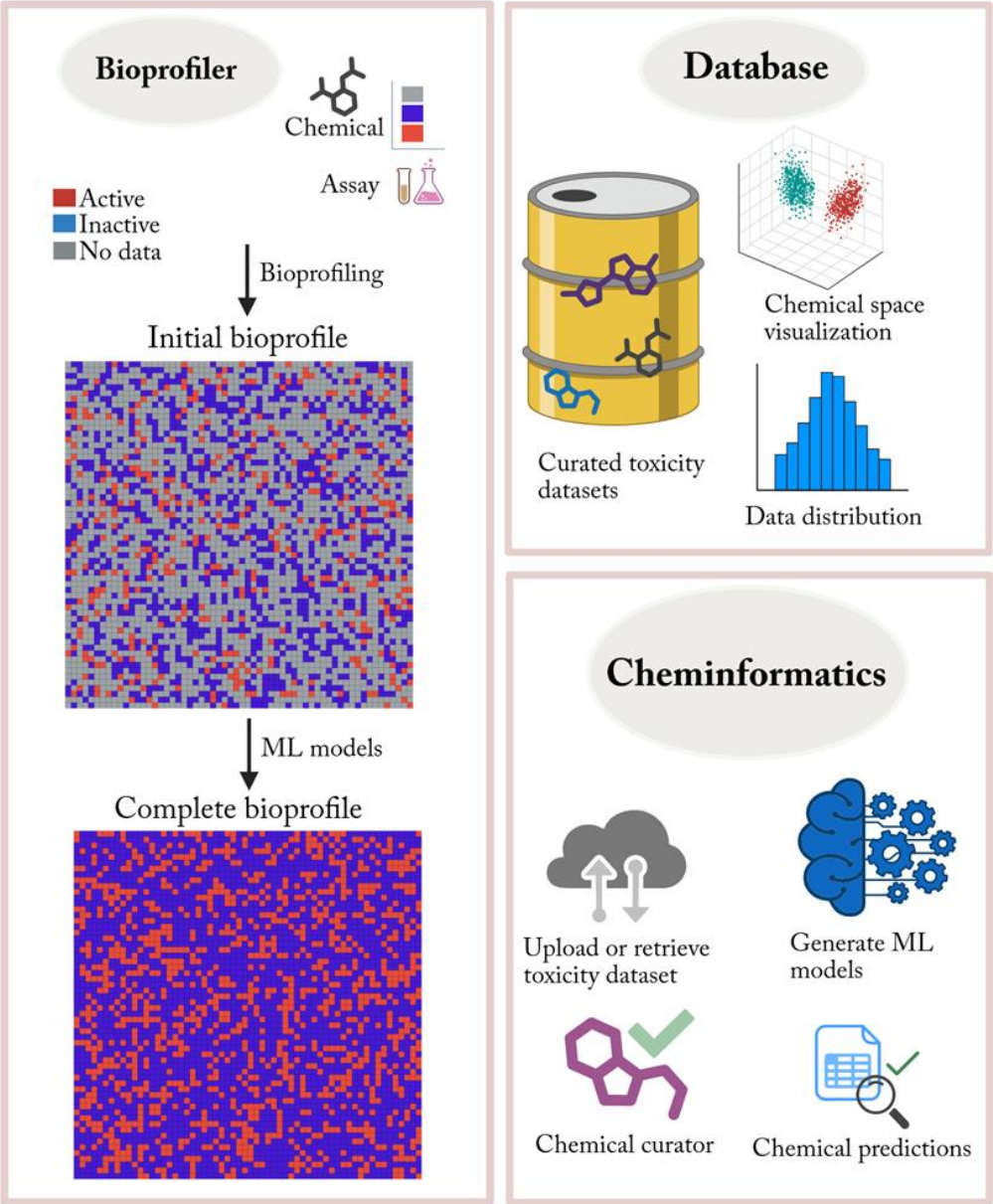
Tutorial

www.toxiverse.com

ToxiVerse



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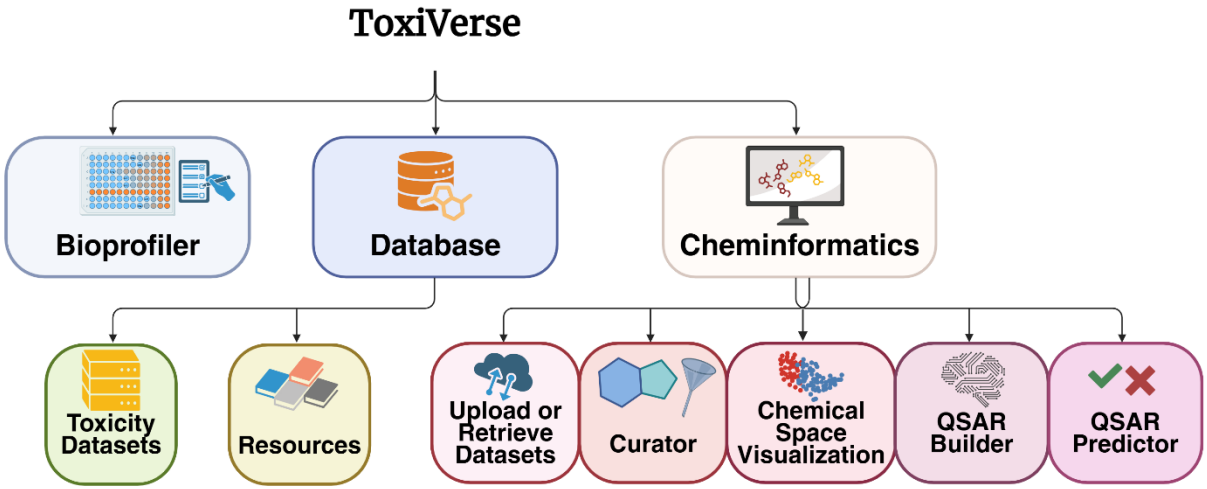
1. Introduction

Computational toxicology plays a significant role in identifying hazardous compounds to protect human health and the environment in a cost-effective manner. A major challenge in this field is the lack of publicly available and user-friendly computational tools that can be used for chemical risk assessment with user-provided dataset, especially by users with limited computational expertise.

To address this need, we developed Toxicology Universe (ToxiVerse), a web portal designed to assist toxicologists, pharmaceutical researchers, and chemists in assessing chemical safety. Please check Figure below for all the available options.



The options in ToxiVerse. They are explained in detail along with step-by-step tutorial in the following pages.

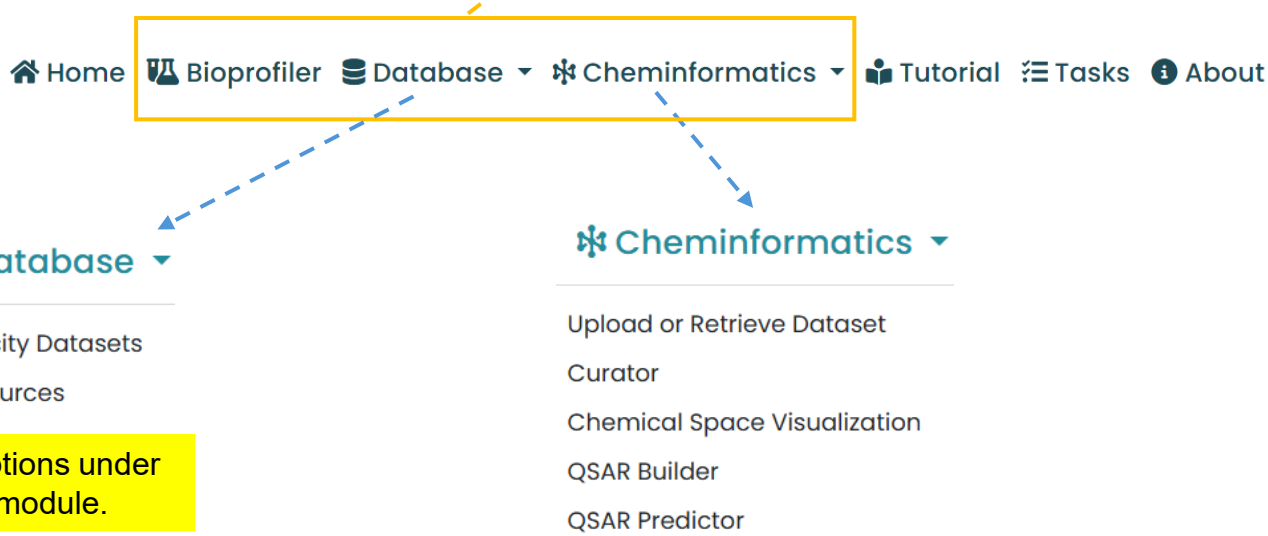
ToxiVerse offers the following modules:

I. **Bioprofiler:** Provides chemical descriptors by profiling PubChem bioactivity results for chemicals of interest, once experimental data gaps are filled using machine-learning models built from key assays.

II. **Database:** Download and visualize curated toxicological datasets. The integrated database includes over 50,000 chemicals across 50 endpoints (mostly related to toxicity), compiled from various sources.

III. **Cheminformatics:** Create QSAR models using either user-uploaded datasets or datasets retrieved from PubChem by providing an Assay ID. A variety of molecular descriptors and machine learning algorithms are supported. Predict toxicity for new chemicals using the models developed. Options for chemical curation and space visualization are also provided.

1. These are the three main modules.



2. Available options under the Database module.

3. Available options under the Cheminformatics module.

Bioprofiler

2. Bioprofiler

Users can upload up to 500 chemicals to profile them using PubChem bioactivity data. Provides chemical descriptors by profiling PubChem bioactivity results for chemicals of interest, once experimental data gaps are filled using machine-learning models built from all 35 key assays if they have at least 100 actives and inactive chemicals.

*A sample file for Bioprofiler looks like below with the list of PubChem CIDs.

4091
2548
5417
1254
1493
1981
14403
2141
3061
6049
3467

1. Upload a file with PubChem Compound IDs eg: sample_dataset.txt

*Sample files to download.

Upload a file containing PubChem CIDs (one per line) to profile by **bioactivity data**. Accepted formats: `.txt` or `.csv`, single column.

If you have other identifiers (e.g., SMILES), use PubChem's batch converter: ID Exchange

Download Sample CSV | Download Sample TXT

Upload Dataset File (Formats: .txt, .csv; max 500 CIDs)

Choose File No file chosen

File must be a single column of CIDs. Empty lines are ignored.

After submission, you will be redirected to a page with your Job ID. It may take 20 minutes on average to complete. The link will be accessible for 24 hours from submission and can be bookmarked.

Build Bioprofile → 2. Click here to start

Bioprofiler Job Status

Job ID

ef6515c1e3cf40f29948bf5e54c667b6

Status

Completed

Submitted

2025-12-08 14:03:26 CST

Completed: 2025-12-08 14:18:18 CST

Job completed successfully! Your results are ready for download below.

Download Results

Initial Bioprofile | Heatmap | Model Metrics | Metrics Plot | Complete Bioprofile

Results to download

What Each Download Option Means

Initial Bioprofile: Exports the bioprofile matrix with chemicals as rows and bioassays as columns. Activity values are encoded as 1 (Active/Probe), -1 (Inactive), and 0 (Inconclusive/Unspecified/empty).

Heatmap: A clustered heatmap of the bioprofile matrix to spot activity patterns across assays and chemicals. Supports up to 2000 assay columns.

Model Metrics: Detailed performance statistics (accuracy, precision, recall, F1-score, ROC AUC) for Random Forest models on top informative assays.

Metrics Plot: Box plot summarizing model performance for quick comparison.

Complete Bioprofile: Returns the complete bioprofile matrix with missing bioactivity values imputed using trained models.

This link and all results will expire 24 hours after submission.

•**Initial Bioprofile**: Exports the bioprofile matrix with chemicals as rows and bioassays as columns. Activity values are encoded as 1 (Active/Probe), -1 (Inactive), and 0 (Inconclusive/Unspecified/empty).

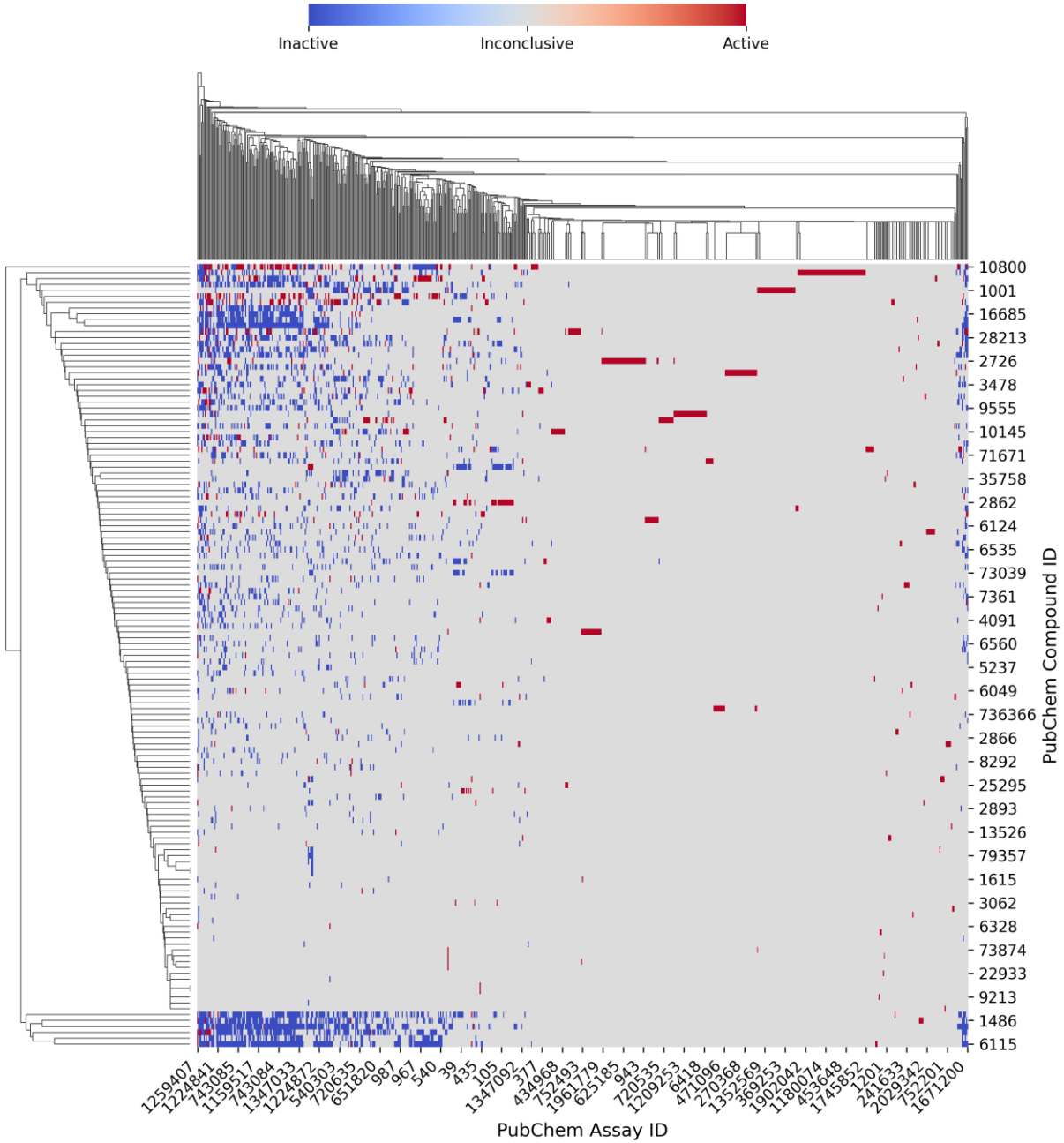
Assays

Chemicals

CID	1224840	1224841	1224868	1259247	1259367	1259369	1259380	1259408
33	0	0	0	0	0	0	0	1
34	-1	0	0	0	0	0	0	0
300	0	-1	0	-1	0	0	0	-1
311	0	0	0	-1	0	0	0	0
611	0	0	0	0	0	0	0	0
674	0	0	0	0	0	0	-1	0
713	0	0	0	0	-1	0	0	0
774	0	-1	0	0	0	0	0	0
887	0	0	0	0	0	0	0	0
991	0	0	0	1	0	0	0	0
1001	0	0	0	0	0	0	0	0
1031	0	0	0	0	0	0	-1	0
1133	0	-1	-1	0	-1	0	-1	0
1140	0	-1	-1	0	0	0	0	-1
1254	0	0	0	0	0	0	0	0
1486	-1	-1	-1	-1	-1	-1	0	1
1493	0	0	0	0	0	0	0	-1
1614	0	0	0	0	0	0	0	0
1615	0	0	0	0	0	0	0	0
1981	0	0	0	0	-1	0	0	0
2039	0	0	0	0	0	0	0	0
2132	0	0	0	0	0	0	0	0
2141	0	0	-1	0	0	0	0	0
2268	0	0	0	0	0	0	0	-1

The values of inconclusive data (0s) circled in blue will be replaced with either 1 or -1 in the Complete Bioprofile.

•**Heatmap**: A clustered heatmap of the bioprofile matrix to spot activity patterns across assays and chemicals. Supports up to 2000 assay columns.

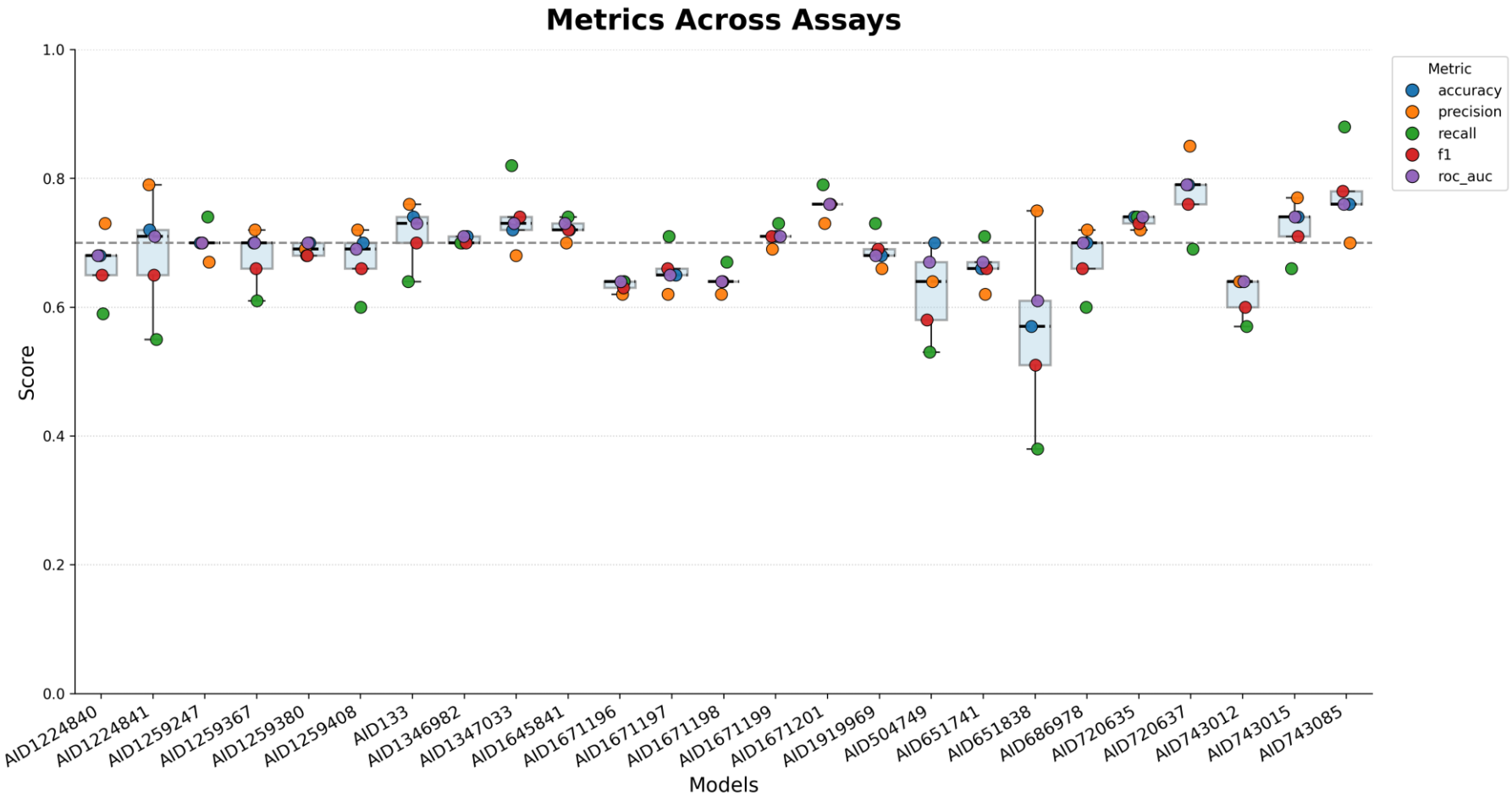


•**Model Metrics**: Detailed performance statistics (accuracy, precision, recall, F1-score, ROC AUC) for Random Forest models on top informative assays.

Model	accuracy	precision	recall	f1	roc_auc
AID1224840_rf_model	0.68	0.73	0.59	0.65	0.68
AID1224841_rf_model	0.72	0.79	0.55	0.65	0.71
AID1224868_rf_model	0.7	0.68	0.7	0.69	0.7
AID1259247_rf_model	0.7	0.67	0.74	0.7	0.7
AID1259367_rf_model	0.7	0.72	0.61	0.66	0.7
AID1259380_rf_model	0.7	0.69	0.68	0.68	0.7
AID1259408_rf_model	0.7	0.72	0.6	0.66	0.69
AID133_rf_model	0.74	0.76	0.64	0.7	0.73
AID1346982_rf_model	0.71	0.7	0.7	0.7	0.71
AID1347033_rf_model	0.72	0.68	0.82	0.74	0.73
AID1645841_rf_model	0.72	0.7	0.74	0.72	0.73
AID1671196_rf_model	0.64	0.62	0.64	0.63	0.64
AID1671197_rf_model	0.65	0.62	0.71	0.66	0.65
AID1671198_rf_model	0.64	0.62	0.67	0.64	0.64
AID1671199_rf_model	0.71	0.69	0.73	0.71	0.71
AID1671201_rf_model	0.76	0.73	0.79	0.76	0.76
AID1919969_rf_model	0.68	0.66	0.73	0.69	0.68
AID504749_rf_model	0.7	0.64	0.53	0.58	0.67
AID651741_rf_model	0.66	0.62	0.71	0.66	0.67
AID651838_rf_model	0.57	0.75	0.38	0.51	0.61
AID686978_rf_model	0.7	0.72	0.6	0.66	0.7
AID720635_rf_model	0.74	0.72	0.74	0.73	0.74
AID720637_rf_model	0.79	0.85	0.69	0.76	0.79
AID743012_rf_model	0.64	0.64	0.57	0.6	0.64
AID743015_rf_model	0.74	0.77	0.66	0.71	0.74

The top assays were selected through Mutual Information (MI) score. MI measures how informative assay outcomes are about the overall activity of chemicals. First, the bioactivity data collected from PubChem assays for target chemicals were transformed into a chemical-bioactivity matrix (i.e. initial bioprofile), where active values were encoded as 1 and inactive or inconclusive values were encoded as 0. Next, each chemical was assigned a binary overall activity label, defined as active if it was active in at least one assay. Finally, the MI between the activity outcome of each assay and the overall activity label was calculated. An assay achieves a high MI score when both its actives (1) and inactives (0) closely match the overall activity labels of as many chemicals as possible. The more consistently an assay distinguishes overall active from inactive chemicals, the higher its MI is. From each assay, a maximum of 500 active and 500 inactive chemicals were collected to build QSAR model, with a minimum of 100 chemicals in both classes required for the assay to be selected.

•**Metrics Plot**: Box plot summarizing model performance for quick comparison.



•**Complete Bioprofile**: Returns the complete bioprofile matrix with missing bioactivity values imputed using trained models.

Assays

Chemicals

CID	1224840	1224841	1224868	1259247	1259367	1259380	1259408
33	-1	-1	-1	-1	-1	-1	1
34	-1	-1	-1	-1	-1	-1	1
300	-1	-1	-1	-1	-1	-1	-1
311	-1	-1	-1	-1	-1	-1	-1
611	-1	-1	-1	-1	-1	-1	-1
674	-1	-1	-1	-1	-1	-1	-1
713	-1	-1	-1	-1	-1	-1	1
774	-1	-1	-1	-1	-1	-1	-1
887	-1	-1	-1	-1	-1	-1	-1
991	1	1	-1	1	-1	-1	-1
1001	-1	-1	-1	-1	-1	-1	-1
1031	-1	-1	-1	-1	-1	-1	-1
1133	-1	-1	-1	-1	-1	-1	-1
1140	-1	-1	-1	-1	-1	-1	-1
1254	-1	-1	-1	-1	-1	-1	-1
1486	-1	-1	-1	-1	-1	1	1
1493	1	1	-1	1	1	-1	-1
1614	-1	-1	-1	1	-1	-1	1
1615	-1	-1	-1	1	-1	-1	1
1981	-1	1	-1	-1	-1	1	1
2039	-1	-1	-1	-1	-1	-1	-1
2132	-1	-1	-1	-1	-1	-1	1
2141	-1	-1	-1	-1	-1	-1	-1
2268	-1	1	1	1	-1	-1	-1

Some of the complete bioprofile values (circled in red) are replaced with either 1 or -1 in the Complete Bioprofile that were earlier 0s in Initial Bioprofile .

Database

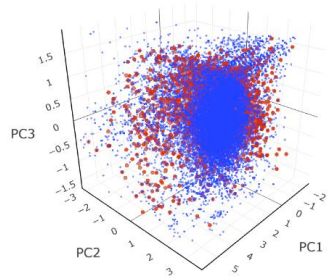
Toxicity Datasets

3. Toxicity Datasets

Allows users to download and visualize curated toxicological datasets, including chemical space, endpoint distributions, and provides relevant bioassays from PubChem for the selected endpoints. The dataset contains over 50,000 chemical records across 50 endpoints mainly related to toxicity, collected from various sources.

Select an endpoint in a toxicity dataset to display PCA plot, histogram and Relevant bioassays.

Principal Component Analysis

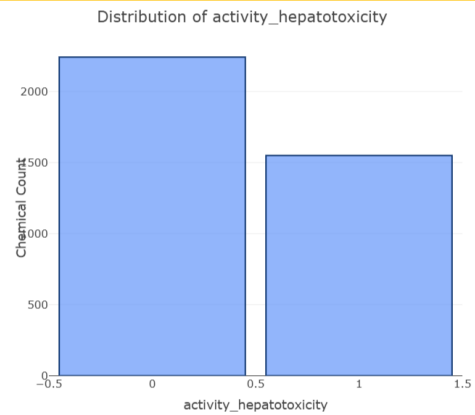


- Compounds with endpoint data in this dataset
- Compounds without endpoint data in this dataset
- Compounds from other datasets

Endpoint Distribution

Select endpoint:

activity_hepatotoxicity (Hepatotoxicity_curated)



Relevant Bioassays

AID	Bioassay Name	Inactive	Inconclusive	Active	Active Rate
651613	Luminescence-based cell-based high throughput confirmation assay for inverse agonists of the liver receptor homolog-1 (LRH-1; NR5A2)	8	0	58	0.88
651614	Counterscreen for inverse agonists of the liver receptor homolog-1 (LRH-1; NR5A2): Luminescence-based cell-based high throughput assay to identify inverse agonists of the Steroidogenic Factor 1 Nuclear Receptor (SF1; NR5A1)	11	0	55	0.83
651611	Counterscreen for inverse agonists of the liver receptor homolog-1 (LRH-1; NR5A2): Luminescence-based cell-based high throughput assay to identify nonselective inhibitors of the Steroidogenic acute regulatory protein (StAR) promoter or luminescence assay artifacts	25	0	41	0.62

Bioassays are ranked by active rate (Active / (Active + Inactive)) for the selected endpoint. Assays with ≥1 Active are included. To stabilize small-sample assays, we use Bayesian-adjusted scores and retain up to the top 500 assays per endpoint. Inconclusive results are shown but excluded from calculations.

Download a Database

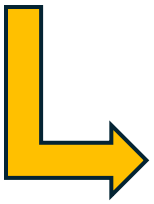
Select database:

Hepatotoxicity_curated

Download

Exports the selected dataset to a CSV file.

Database



Resources

4. Resources
The details of the datasets available including references.

First few rows of the Resources Table

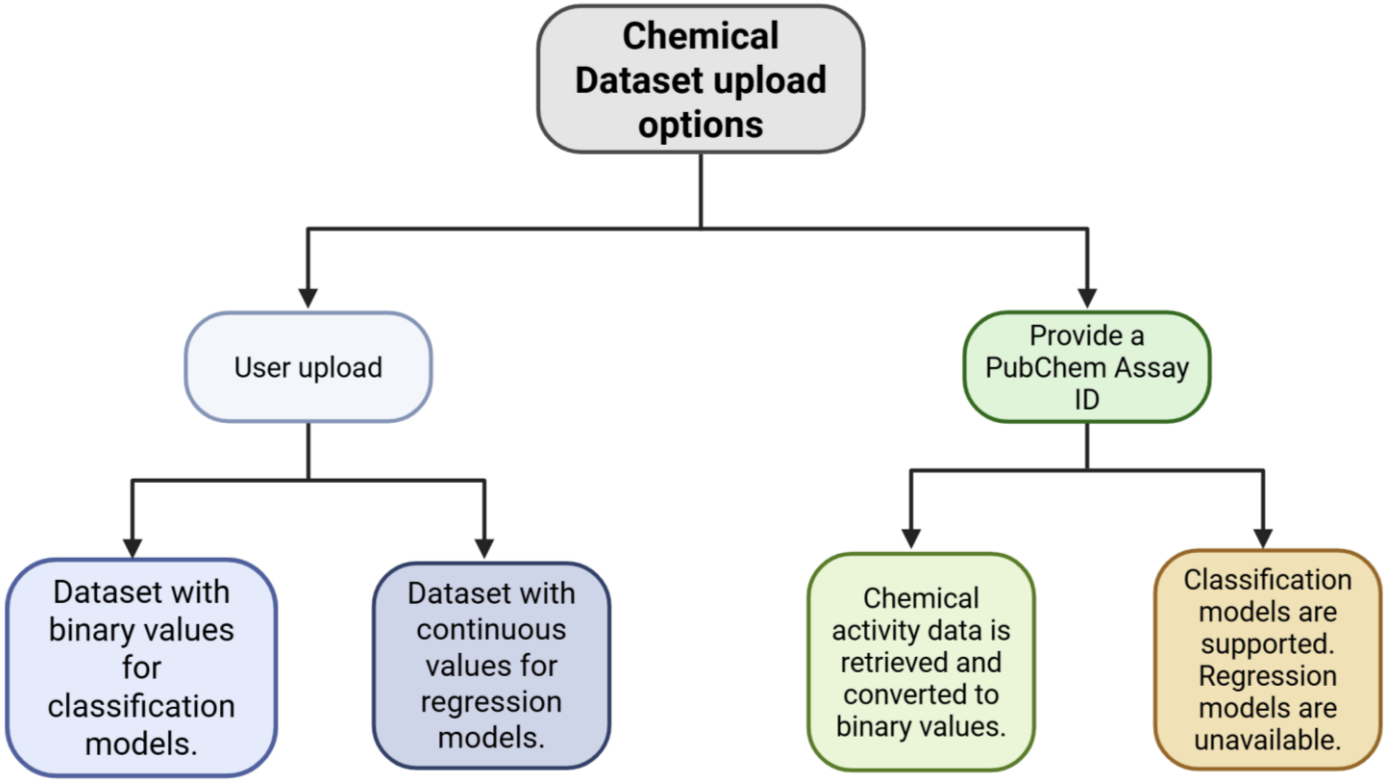
Hyperlinks

Curated Datasets

Dataset Name	Total # of Compounds	Data Type	Dataset Source	Dataset Description
Acute toxicity	7332	log10 mol/kg-bw	Zhu et al.	Used publicly available rat oral acute toxicity data, with LD_{50} (Rat oral) values classified into toxicity categories (e.g., high, moderate, low) according to Globally Harmonized System (GHS) thresholds.
Aquatic toxicity	675	-log10 of Conc. (μmol/L)	Klopman et al.	Collected from standardized 96-hour LC_{50} test data for Pimephales promelas (fathead minnow), sourced from the EPA's ECOTOX database and additional public toxicology resources.
BBB (Blood Brain Barrier)	438	logBB	Wang et al.	Compounds with experimental logBB values was compiled and curated using ChemAxon and CASE Ultra tools.
Bioavailability	1141	oral bioavailability (%F)	Kim et al. Moda et al.	Compiled across public and literature sources. Chemical structures were standardized, and %F values were harmonized to resolve discrepancies.
Carcinogenicity	342	Binary, 0=Non-Carcinogen; 1=Carcinogen	Chung et al.	342 unique organic compounds from the EPA's IRIS database, labeled as carcinogenic or noncarcinogenic based on oral slope factor (OSF), a quantitative measure for oral cancer risk.

Cheminformatics

Upload or Retrieve dataset



5. Upload or Retrieve datasets

Data can be in the Comma-Separated Values (CSV) or Structure Data Format (SDF) format to upload. Sample files provided. You can upload or retrieve up to 1000 chemicals. Instead of uploading a dataset, you may also import chemicals with structure-activity information from PubChem by entering the PubChem Assay Identifier (AID).

1. Select a format

2. Upload a file (eg: sample_dataset.csv).

Please check the Sample files provided.

3. eg: CID

*Accepts up to 1000 chemicals.

Upload a dataset

Choose a file type to upload a dataset. For CSV, include a SMIL column and enter column nam exactly as they appear (case-sensitive).

Choose file format:

- (•) CSV [Download Sample CSV]
- ( ) SDF [Download Sample SDF]

4. eg: Activity_Binary.

Select file

[Choose File] No file chosen

Compound ID (e.g., CID; must be a property in SDF or a column in CSV)

CID

5. eg: SMILES.

Activity column (e.g., Activity_Binary or Activity_Reg)

Activity_Binary

SMILES column (CSV only) (e.g., SMILES)

SMILES

Select dataset type

- (•) Binary
- ( ) Continuous

[Upload dataset]

6. Choose dataset type and click on Upload dataset.

A sample file for Upload Dataset looks like this.

CID	SMILES	Activity_Binary
177	CC=O	1
299	O=C1C2(Cl)C3(Cl)C4(Cl)C(Cl)(Cl)C5(Cl)C3(Cl)C1(Cl)C5(Cl)C24Cl	1
546	O=C(O)CP(=O)(O)O	0
564	NCCCCCC(=O)O	0
774	NCCc1cnc[nH]1	0
785	Oc1ccc(O)cc1	0
864	O=C(O)CCCCC1CCSS1	0
896	COc1ccc2[nH]cc(CCNC(C)=O)c2c1	0
938	O=C(O)c1cccnc1	0
992	Oc1c(Cl)c(Cl)c(Cl)c(Cl)c1Cl	1
1046	NC(=O)c1cnccn1	0
1054	Cc1ncc(CO)c(CO)c1O	0
1123	NCCS(=O)(=O)O	0
1318	c1cnc2c(c1)ccc1cccnc12	1
1355	Clc1cccc(N2CCNCC2)c1	0
1359	FC(F)(F)c1ccccc1-n1ccnc1	0
1483	Oc1c(Br)cc(Br)cc1Br	1
1674	Cc1ccc2cc3c(ccc4ccccc43)c3c2c1CC3	1
1727	Nc1ccncc1	0

Uploaded file is displayed here

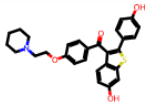
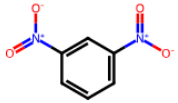
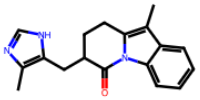
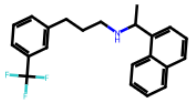
Select dataset:

sample_classification

Preview

Show 10 entries

Search:

Chemical	Activity	Structure
5035	1	
7452	1	
180494	1	
7430	1	
156418	1	

Showing 1 to 10 of 99 entries

Previous 1 2 3 4 5 ... 10 Next

Remove dataset

Download dataset as CSV

Download or remove the selected dataset

Import a PubChem Bioassay

Instead of uploading a file, pull compounds from PubChem by entering an Assay ID (AID). Once imported, the dataset will show up in the "Select dataset" list.

Enter PubChem AID (e.g., 1000)

AID

Import dataset

1. Enter an AID and click to retrieve the dataset from PubChem data stored in local database.

*Retrieves up to 1000 random chemicals. 500 actives and 500 inactives.

Cheminformatics

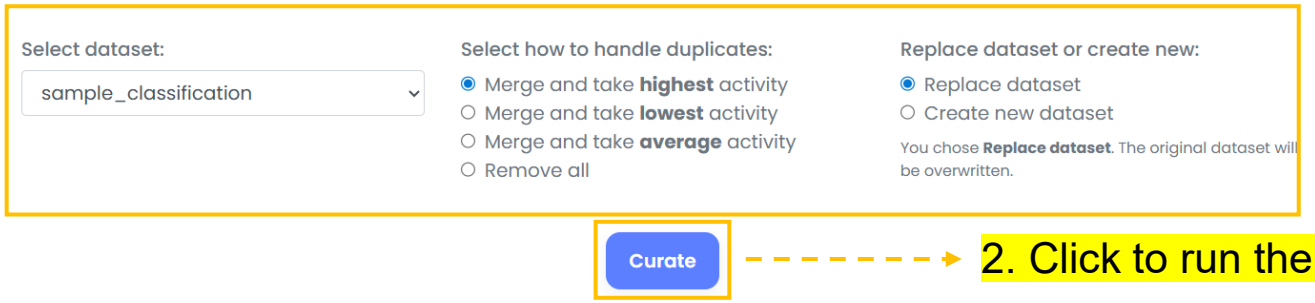
Curator

6. Curator

Cleans the chemical structures and prepares them for next steps such as model generation by the following steps:

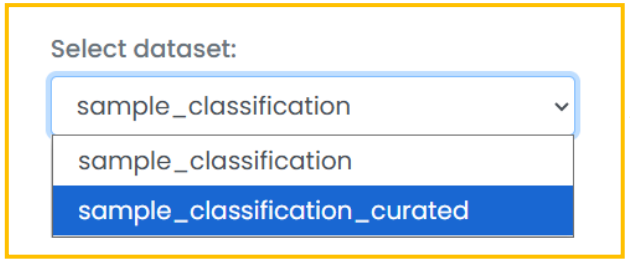
i. Check and clean chemical structures.
ii. Standardize chemical structure representation (e.g, updating valencies, removing charges, etc.).
iii. Strip salts and solvents, remove mixtures by retaining the parent non-salt/non-solvent component.
iv. Merge or remove duplicated structures.

1. Select your dataset and options.



2. Click to run the curation.

Curation complete. The curated dataset is now available in the *Select Dataset* dropdown below. You can also access it in Cheminformatics → Chemical Space Visualization and Cheminformatics → QSAR Builder.



A file named sample_classification_curated will be generated if ‘Create new dataset’ option was chosen above instead of ‘Replace dataset’.

Cheminformatics

Chemical Space Visualization

7. Chemical Space Visualization

Principal Component Analysis (PCA) is a dimension reduction technique used for visualizing chemical space.

1. Select your dataset.

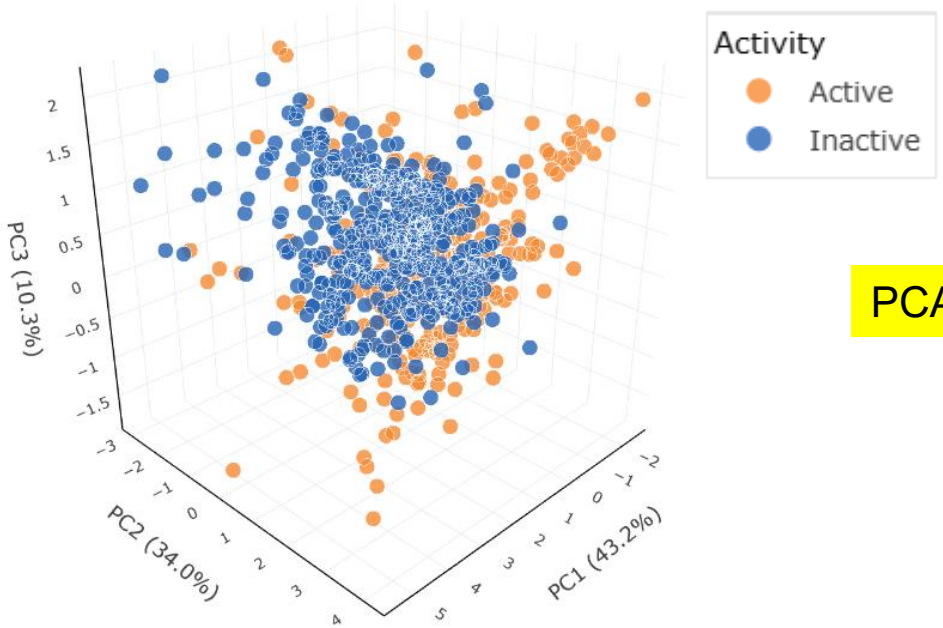
Select dataset:

sample_classification

Perform PCA

2. Click to generate PCA plot.

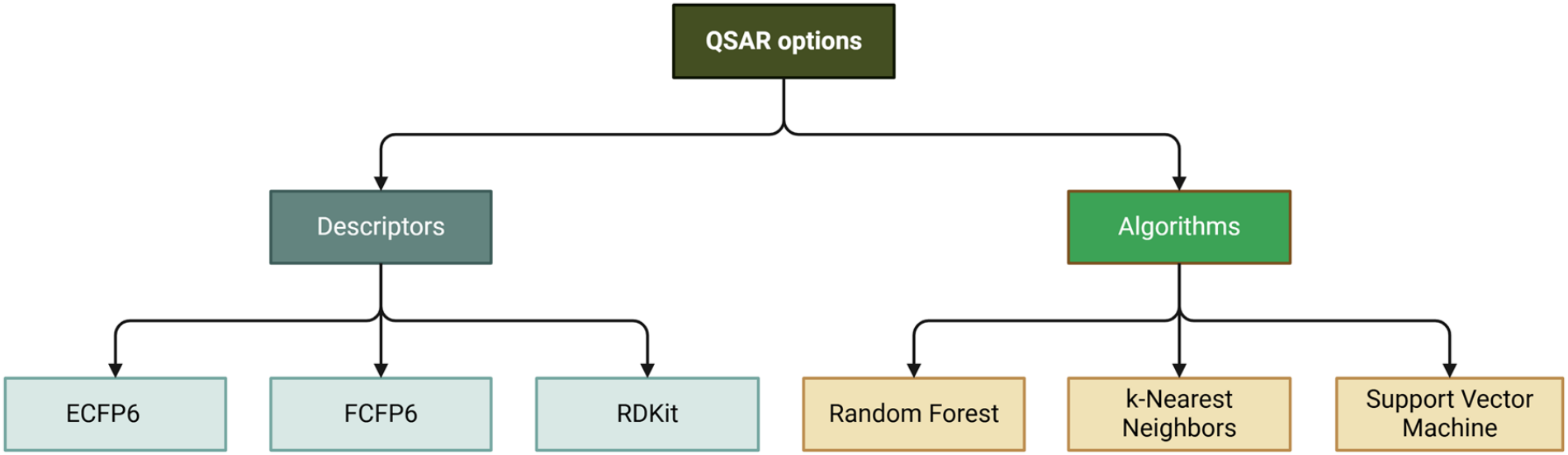
PCA plot is generated for dataset sample_classification and displayed below.



PCA figure is generated.

Cheminformatics

QSAR Builder



8. QSAR Builder

Create QSAR models using a variety of descriptors and algorithms with either a user-uploaded dataset or a dataset obtained by providing a PubChem Assay ID.

1. Choose your dataset

2. Choose Features, Algorithms and Activity type to build QSAR models. Multiple options can be chosen

Select dataset(s):

sample_classification ×

You may select one or more datasets.

Select features:

- [x] ECFP6
- [ ] FCFP6
- [ ] RDKit Descriptors

Select algorithm:

- [x] Random Forest (RF)
- [x] k-Nearest Neighbors (kNN)
- [ ] Support Vector Machine (SVM)

Select activity type:

- (•) Classification
- ( ) Regression

After submission, you will be redirected to a page with your Job ID. It may take 20 minutes on average to complete. The link will be accessible for 24 hours from submission and can be bookmarked.

Build QSAR

3. Click to generate model(s)

Model training complete. Performance metrics are now visible in the *Trained Models* table below. View your results at: http://toxiverse.com/job/5762f2ac64354fe8a1acb595b8b2e454

Download Results

Trained Models (2)

Classification metrics file | Classification metrics figure

Model	Type	Accuracy	F1 Score	Precision	Recall	ROC AUC
Classification-ECFP6-kNN-Classification kNN ECFP6	Classification	0.71	0.73	0.67	0.81	0.80
Classification-ECFP6-RF-Classification RF ECFP6	Classification	0.74	0.73	0.76	0.71	0.82

4. Check the evaluation metrics of your model along with link to access the generated models.

→Use Your Model for Predictions

Your trained model(s) are ready! You can now use them to predict activity/toxicity for new chemicals.

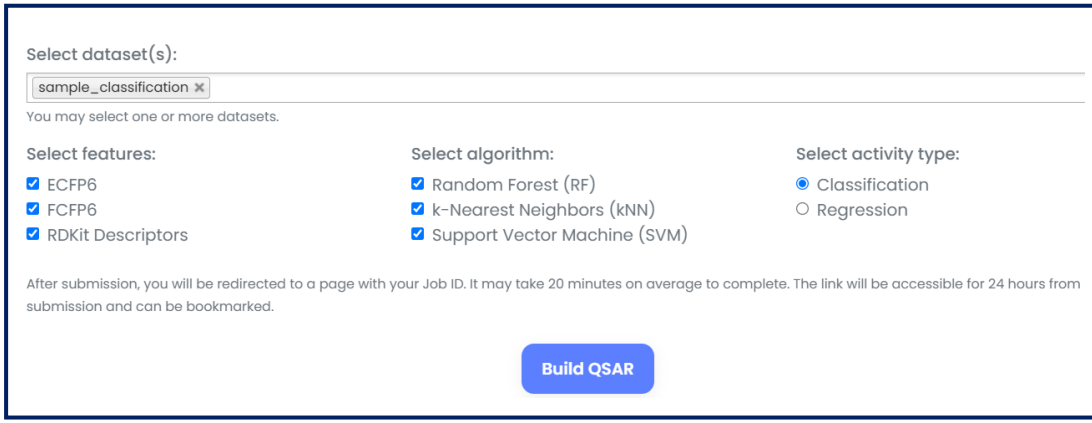
Go to QSAR Predictor

Note: Opens in new tab with models from this session.

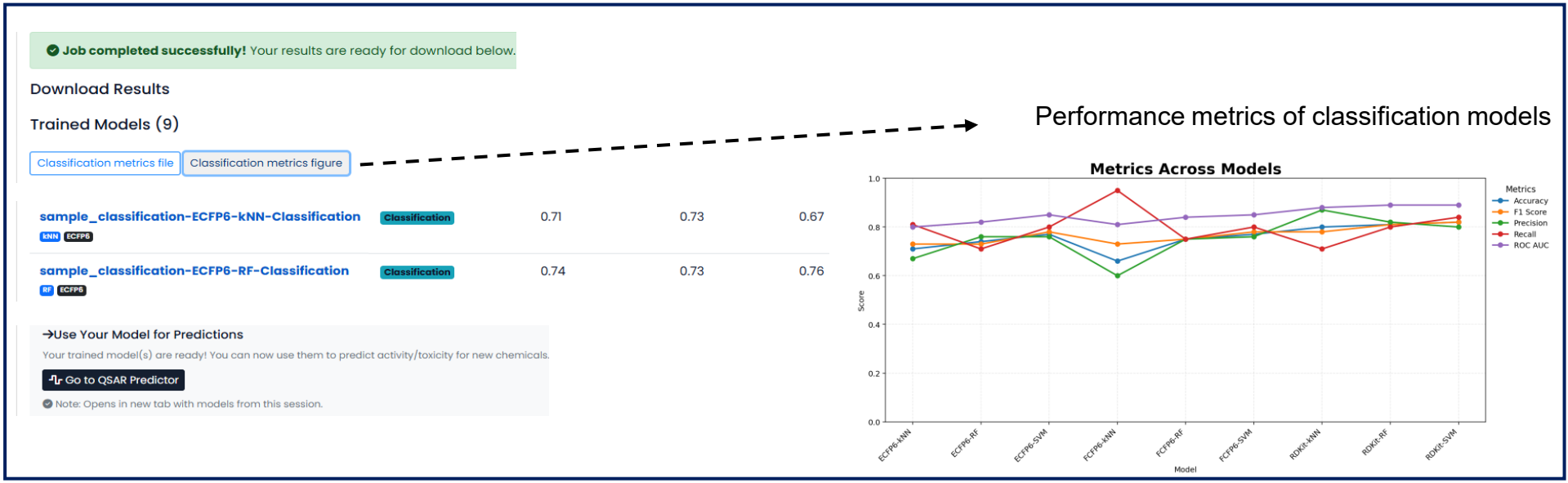
5. Directs to access the generated models.

- If all the fingerprints and algorithms are selected like below, then the performance metrics look like the line plot below.

- *Classification metrics figure* option below is used to download plot and a csv format of those scores can be obtained from *Classification metrics file* below.



Dataset selection and multiple options for QSAR model generation



Performance metrics of classification models

ToxiVerse

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Tasks

Monitor the status of background jobs for chemical curation and QSAR model building. This table refreshes automatically every 5 seconds.

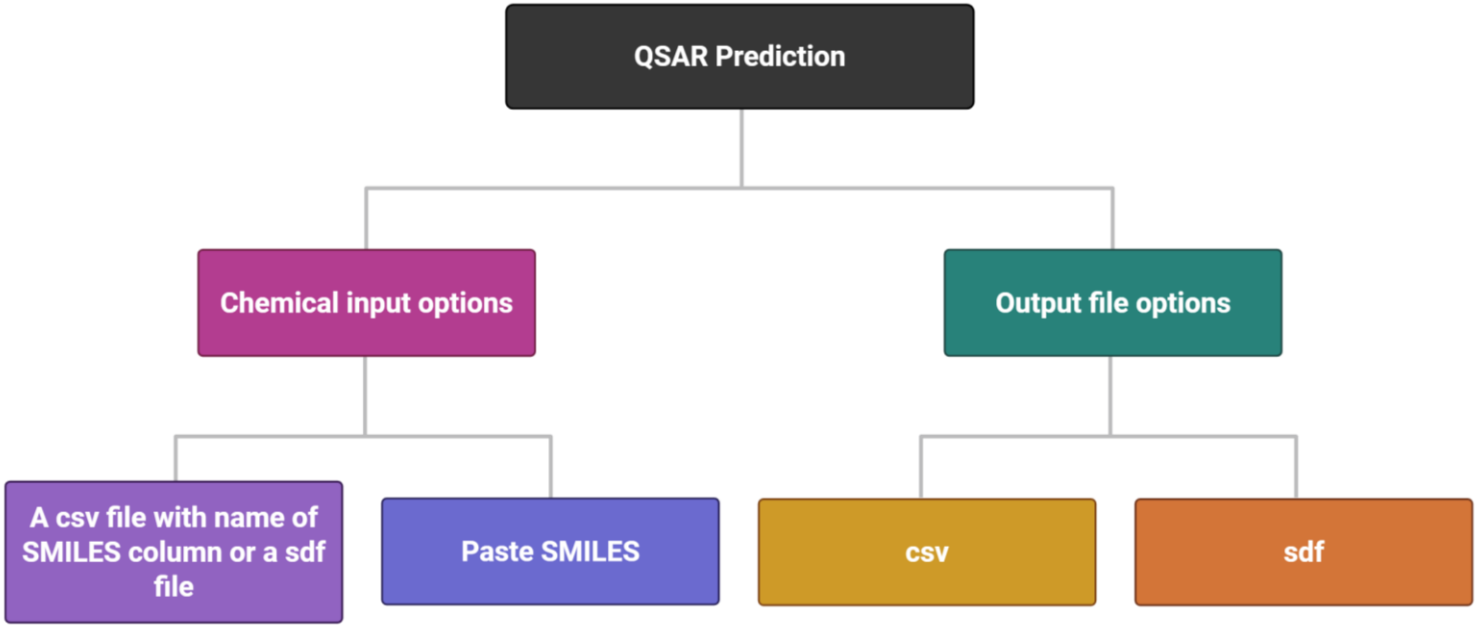
Task Queue

Description	Status
Building QSAR Model on sample_classification&ECFP6&RF&Classification	Complete
Building QSAR Model on AID_1000_curated&ECFP6&RF&Classification	Complete
Curating AID_1000_curated chemicals	Complete
Curating AID_1000 chemicals	Complete
Importing structure-activity data for AID 1000: Bioassay 1000	Complete

You can monitor the status, including live updates, of ***Cheminformatics*** module jobs here, except for the QSAR Predictor option which provides instant result.

Cheminformatics

QSAR Predictor



9. QSAR Predictor

Predict toxicity for new chemicals using models developed in QSAR Builder option.

*Accepts up to 100 chemicals.
*Sample files can be downloaded.

1. Select your QSAR model.

*Multiple QSAR models can also be selected.

Select QSAR model:

sample_classification-ECFP6-RF-Classification

Select CSV or SDF File:

Choose File No file chosen

Predict

SMILES column name in the CSV (default is SMILES if left empty).

SMILES

Input Method:

- Upload CSV/SDF File
- Paste SMILES

Select Output File Format:

- CSV
- SDF

2. Select an input and output format.

3-1. Choose a file eg: Prediction_dataset.csv and enter eg: SMILES, then click on Predict.

3-2. Paste SMILES and click Predict

Input Method:

- Upload CSV/SDF File
- Paste SMILES

Paste up to 100 SMILES (one per line):

CCO\nC1=CC=CC=C1\n...

Paste Sample SMILES Predict

Select Output File Format:

- CSV
- SDF

Prediction of new chemicals using a selected model with corresponding output scores.

Prediction results with "_Prediction" added to the model name

SMILES	sample_classification-ECFP6-RF-Classification_Prediction
C1=CC(=CC=C1C(=C(Cl)Cl)C2=CC=C(C=C2)Cl)Cl	1
C1=CC(=C(C=C1Cl)Cl)Cl	1
Cn1c(=O)c2c(ncn2CCO)n(C)c1=O	0
Nc1c2c(nc3ccccc13)CCCC2	0
COc1ccc2c(c1)c(CC(=O)OCC(=O)O)c(C)n2C(=O)c1ccc(Cl)cc1	0
CC(=O)Nc1ccc(O)cc1	0
CC(=O)Nc1nnc(S(N)(=O)=O)s1	0
CCOCN(C(=O)CCl)c1c(C)cccc1CC	1
CC(=O)c1ccc(S(=O)(=O)NC(=O)NC2CCCCC2)cc1	0
N#CC(C#N)=Cc1ccc(O)c(O)c1	0
CCc1cccc(CC)c1N(COC)C(=O)CCl	1

10. Contact us

Rowan University: 201 Mullica Hill Rd, Robinson Hall, Glassboro, NJ 08028
Tulane University: Hutchinson Memorial Building (School of Medicine), 1415 Tulane Ave, New Orleans, LA 70112
Questions, comments, and general inquiries can be emailed to toxiverse.help@gmail.com.

11. About us

The Zhu Lab uses cheminformatics algorithms, workflows, and other computational tools to model chemical toxicity, ADME (Absorption, Distribution, Metabolism, and Excretion), and other biological activities. These models support regulatory chemical toxicity assessments and the computer-aided drug discovery (CADD) process.

ZHU LAB

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